

Swaps Rates (semi 30/360 versus 3-month LIBOR)		
	Bid	Offer
1 Year	5.00%	5.02%
2 Year	5.10%	5.12%
3 Year	5.16%	5.18%
4 Year	5.20%	5.22%
5 Year	5.25%	5.27%

In the questions that follow, assume that the dealer quotes levels using the relevant bids and offers given above.

a) Given the swap rates above, what are the 5-year bid and offer swap rates on a quarterly money basis, assuming the floating side remains 3-month LIBOR quarterly money?

b) Given the swap rates above, what are the 5-year bid and offer swap rates on a monthly money basis, assuming the floating side remains 3-month LIBOR quarterly money?

c) Given the swap rates above, what are the 5-year bid and offer swap rates on a monthly money basis, assuming the floating side is 1-month LIBOR monthly money?

d) Suppose that a client wants to enter into a swap with the dealer in which the client will pay 1-month LIBOR – 25 bps monthly Act/360 for five years. What fixed rate will the swap dealer quote assuming the fixed rate is also paid on a monthly Act/360 basis?

e) Suppose that a client wants to enter into a swap with the dealer in which the client will receive Prime quarterly Act/360 for five years. What fixed rate will the swap dealer quote assuming the fixed rate is also paid on a quarterly Act/360 basis?

f) Suppose that a client wants to enter into a swap with the dealer in which the client pays Prime – 250 bps quarterly Act/360 for three years, and the dealer pays 1-month LIBOR + x bps monthly Act/360. What is x ?

4. Suppose that r_{xb} is a rate of interest compounded x times per year, and r_{yb} is a rate of interest compounded y times per year. An investor can borrow or lend money at either of these rates for a year. Show that

$$r_{yb} = y \times \left[\left(1 + \frac{r_{xb}}{x} \right)^{x/y} - 1 \right],$$

which is the motivation behind the approximation formula in Equation 1.8.

5. Referring to Table 1.1, the convention is that the 12-year swap spread (bid or offer) is quoted as a spread to the mid-market yield of the on-the-run 10-year Treasury. Suppose instead that the 12-year swap spread were quoted as a spread to the (mid-market) interpolated 10-year and 30-year Treasury yields. In this case, how should the bid and offer 12-year spread be quoted in order to come up with the same bid and offer side 12-year swap rates, as in Table 1.1?

Chapter 2

The Risk Characteristics and the Traditional Uses of Swaps

In this chapter we will consider two things: the risks inherent when one enters into a swap and the traditional ways in which investors use swaps. When an investor enters into a swap, there are several sources of risk that he faces. In particular, there are three primary sources of risk in a fixed-floating swap: interest rate risk, spread risk, and counterparty risk. Cross-currency swaps also have currency risk. We shall discuss each of these in this chapter.

It would seem an unlikely pairing, then, to discuss some of the traditional ways in which investors have used swaps in this chapter. But it is precisely these sources of risk (interest rate risk, spread risk, and currency risk, but not counterparty risk) that are the reasons that investors put on these swaps in the first place. Movements in interest rates, swap spreads, and/or exchange rates will change the value of a swap, which makes swaps an effective hedge for many assets and liabilities.

2.1 Interest Rate Risk

As we mentioned in Chapter 1 (and will discuss in great detail in Chapter 3), the fixed rate in a fixed-floating swap is set in such a way when the trade is executed that the swap has a net present value of zero (which we will denote $NPV = 0$). Although a fixed-floating swap is an $NPV = 0$ transaction

Maturity	Swap Rate
Year 1	5.00%
Year 2	5.11%
Year 3	5.30%
Year 4	5.47%
Year 5	5.60%
Year 7	5.70%
Year 10	5.90%
Year 30	6.20%

Table 2.1: A Sample Swap Curve

when first executed, its value changes as swap rates move.¹ Let us now consider how the value of \$100mm swaps of various maturities changes with an instantaneous parallel shift in the *swap curve* shown in Table 2.1. The swap curve refers to the relationship between maturity and swap rates. It is analogous to the Treasury yield curve, with the exception that swap rates replace Treasury yields. The interest rate risk inherent in these swaps is demonstrated in Table 2.2.

These numbers were computed using a proprietary swap pricing model.² It is important to note two things before we look at any of these computations. First, the numerical results presented here depend very much on the particular swap model used; if we were to use a different model, we would likely obtain slightly different results. Different models employ slightly different methodologies and will compute different values for these types of sensitivities. That being said, the differences computed by different models that employ mainstream methods to price swaps are typically pretty small. Second, and more important, the computations presented in Table 2.2 depend very much on the particular swap curve that was in effect at the time. As swap rates move around, these numbers can vary significantly.

Referring to Table 2.2, we see that if we enter into a \$100mm 5-year swap (and the fixed coupon has been set), if there is an instantaneous parallel shift in the swap curve downward 100 bps, then the value of the swap is \$4.185mm in favor of the fixed rate receiver. If instead swap rates were to shift up 100 bps, then the value of the swap is \$3.963mm against the fixed rate receiver.

¹ Actually, in general the value of a swap will change with the passage of time even if swap rates do not move. This characteristic is quantified by a measure known as *roll-down*, which we shall examine further in Chapter 8.

² We will look at developing a swap pricing model of our own in Chapter 3.

Instantaneous Swap Curve Shift				
Maturity of Swap	-200 bps	-100 bps	+100 bps	+200 bps
2 year	+3.338	+1.646	(1.601)	(3.160)
3 year	+5.270	+2.586	(2.492)	(4.894)
4 year	+6.918	+3.379	(3.228)	(6.313)
5 year	+8.607	+4.185	(3.963)	(7.715)
7 year	+11.763	+5.669	(5.275)	(10.184)
10 year	+15.982	+7.604	(6.904)	(13.179)
30 year	+33.017	+14.670	(11.821)	(21.423)

Table 2.2: Value of \$100mm Swap for an Instantaneous Parallel Shift of the Swap Curve in Table 2.1 from the Perspective of Fixed Rate Receiver (\$ millions)

Notice that the fixed rate receiver makes more in a rally than he loses in a sell-off of equal magnitude.

As we alluded to in Chapter 1, the position of the fixed rate receiver is somewhat analogous to someone who owns a typical fixed rate bond. The numbers in Table 2.2 lend further credence to this interpretation; the owner of a fixed rate bond benefits as the market rallies and is hurt as the market sells off. Moreover, for rallies and sell-offs of equal magnitude, the owner of a fixed rate bond makes more in the rally than he is hurt in a sell-off. More formally, these phenomena are quantified in the analysis of fixed rate bonds as modified duration and convexity.³

Consider now a fixed rate bond whose dirty price, P , can be regarded as a function of its yield to maturity. The *modified duration* of the bond, D_m , is given by

$$dP = -D_m \times P \times dy, \quad (2.1)$$

which relates the change in the value of the bond dP to its modified duration D_m , its dirty price P , and a change in its yield to maturity dy . The modified duration of a fixed rate bond is not constant but depends on the level of its yield to maturity. A bond that exhibits positive convexity has the property that its modified duration increases as its yield to maturity decreases. Such a bond exhibits the characteristic that its value increases by more in a rally than it declines in a sell-off of equal magnitude. A bond with positive mod-

³For a brief review of modified duration and convexity, as well as other concepts often employed in fixed income analysis, see Appendix B.

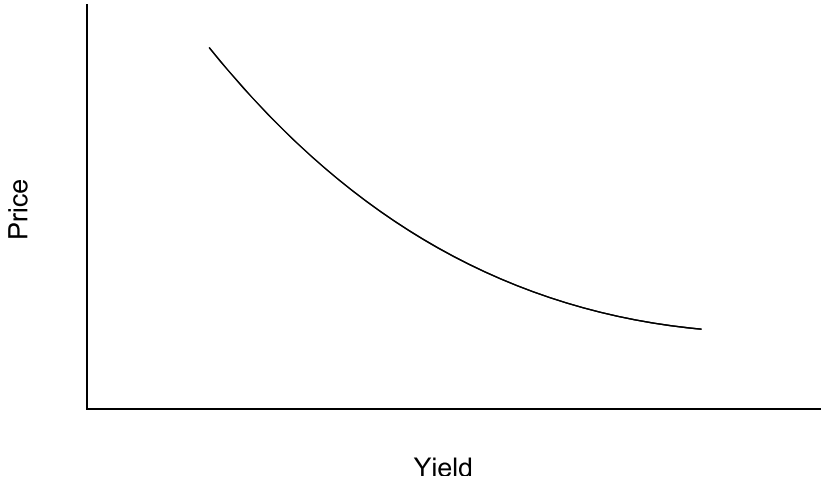


Figure 2.1: Price-Yield Relationship of a Bond with Positive Modified Duration and Positive Convexity

ified duration and positive convexity has a price-yield relationship as shown in Figure 2.1.

However, for swaps it is not common to employ the concept of modified duration when analyzing interest rate sensitivity. There are a couple of reasons for this. First, as we will show in Chapter 3, the valuation of swaps does not lend itself well to the concept of a yield to maturity but rather depends on a sequence of discount factors. This makes it difficult to employ the definition of modified duration as given in Equation 2.1. Second, the definition of duration as given in Equation 2.1 depends on the price of the bond, P . Yet a swap when initiated is an $NPV = 0$ transaction, meaning its “price” is zero. Although it is possible to overcome this second issue, the industry has nonetheless gravitated in a different direction to quantify the interest rate risk inherent in swaps.

2.1.1 PV01

We define the *present value of a basis point*, denoted $PV01$, of a swap to be the change in the value of a swap due to a 1 basis point parallel shift of the swap curve. We will say that a swap has a positive $PV01$ if the value of the swap increases when the swap curve is shifted downward by 1 bp. We will say that a swap has negative $PV01$ if the value of the swap decreases when the swap curve is shifted downward by 1 bp. Thus, a swap in which we

Maturity of Swap	PV01	PV01	
		without LIBOR Set	PV Coupon 01
2 year	\$16,322	\$18,858	\$18,858
3 year	24,882	27,418	27,506
4 year	32,818	35,354	35,628
5 year	40,237	42,773	43,321
7 year	53,063	55,599	57,215
10 year	69,253	71,789	75,007
30 year	112,636	115,172	113,872

Table 2.3: PV01s and PV Coupon 01 in Dollars Given the Swap Curve in Table 2.1. Assumed Notional of \$100mm in Each Case.

are receiving fixed has a positive PV01, and a swap in which we are paying fixed has a negative PV01. The PV01s for swaps of various maturities were computed using a proprietary swap model and are shown in Table 2.3. The column labeled “PV01” provides these PV01 calculations and assumes that the first LIBOR has set in the swap, and thus a parallel shift in the swap curve does not affect this setting. This measure is the most popular measure for quantifying the interest rate sensitivity of swaps.

This table contains another set of interest rate sensitivity numbers that are sometimes employed and that we refer to as “PV01 without LIBOR set.” These calculations (artificially) assume that the initial LIBOR setting in these spot starting swaps has not fixed yet. Thus, under this assumption a parallel shift in the swap curve will affect the initial LIBOR setting. Sometimes people prefer to look at these PV01 numbers because it allows them to isolate the interest rate risk inherent on the fixed side of the swap separately from the risk associated with the floating side. In comparison, a PV01 number (with LIBOR set) presents the net exposure of the fixed side and the floating side, incorporating the fact that the initial LIBOR set creates (de minimis) interest rate sensitivity on the floating side that minimally offsets some of the interest rate sensitivity inherent in the fixed leg of the swap.

In addition to these two metrics of interest rate sensitivity, this table also contains numbers called “PV Coupon 01.” For a given swap curve, these numbers indicate how much the value of the swap changes if we move the fixed coupon in the swap by 1 bp. Here we adopt the convention that a positive (negative) PV Coupon 01 indicates that if we increase the coupon by 1 bp, then the value of the swap increases (decreases) to us. Thus, a positive PV Coupon 01 is associated with a swap in which we are receiving fixed.

PV01 and PV Coupon 01 are two very different things conceptually: the PV01 is calculated by moving the swap curve 1 bp and asking how this changes the value of the swap. The PV Coupon 01 is calculated by holding the swap curve unchanged, moving the fixed coupon 1 bp, and asking how this changes the value of the swap. People sometimes use these two numbers interchangeably, which is definitely incorrect. But as Table 2.3 shows, they are pretty close to one another.

Once again, it is important to note that the numbers in Table 2.3 depend very much on both the particular swap model used and the level of swap rates. In a different rate environment, these numbers would be different. If swap rates were lower, all of these numbers would be higher, and vice versa.⁴

PV01 in bps Upfront Instead of Dollars

People will often equivalently speak about any of these PV01 metrics in terms of bps upfront, instead of in dollars. For example, on \$100mm 10-year swaps, the PV Coupon 01 is \$75,007; \$75,007 is approximately 7.5 bps upfront (i.e., $7.5 \text{ bps} \times \$100\text{mm} \approx \$75,007$). Thus, we will say that 1 bp *running* (meaning a 1 bp change in the coupon of a swap) is worth 7.5 bps upfront. In this way, there is no need to carry around a notional when speaking; one can always apply the relevant notional of the swap to translate into dollars. For example, on a \$50mm 10-year swap, the PV Coupon 01 is also worth 7.5 bps upfront, or $7.5 \text{ bps} \times \$50\text{mm} \approx \$37,504$, half that of a \$100mm swap.

Approximating PV01

The calculation of any of the PV01 numbers we just discussed requires a swap model. In the absence of a model, it is possible to come up with an approximation to the PV Coupon 01 using a financial calculator. To this end, consider an annuity of 1 bp over the life of the swap, discounted using the relevant swap rate. The frequency of the annuity payments should concur with the frequency of the payments on the fixed side of the swap. The present value of this annuity, using the swap rate as the discount rate, is an approximation to the PV Coupon 01. It approximates moving the fixed coupon in the swap by 1 bp and asking how this affects the value of the swap. The approximation manifests itself from using the swap rate to discount, instead of using a sequence of distinct discount factors (which, as we shall discuss in Chapter 3, is most appropriate).

⁴Recall that the duration of a fixed rate bond is not constant but is a function of the level of rates: as rates go lower, duration goes up, and vice versa. An analogous result holds for PV01s of swaps: the lower the swap curve, the higher the PV01 of a swap, and vice versa.

For example, suppose the 10-year semi bond swap rate is 5.90%, as given in Table 2.1. Set

$$\begin{aligned}\text{PMT} &= 1 \text{ bp}/2 \\ n &= 20 \\ i &= 5.90\%/2,\end{aligned}$$

where payment (PMT) represents the annuity of 1 bp paid in semiannual installments, n is the number of semiannual periods in the swap, and i is the swap rate set as discount rate, adjusted for the frequency of the fixed payments. Using a financial calculator, the present value of this annuity is approximately 7.5 bps. On a notional of \$100mm, this PV01 in dollars is

$$\text{\$100mm} \times 7.5 \text{ bps} = \text{\$75,000},$$

very close to the PV Coupon 01 number for a 10-year swap given in Table 2.3.

Examples

Let's work through a few examples to demonstrate how these various PV01 numbers can be used. In each of the following examples, we refer to the PV01 numbers in Table 2.3 that are based on the swap curve in Table 2.1.

- Suppose we enter into a \$100mm 10-year swap and receive at the par swap rate of 5.90%. If the swap curve were to rally 1 bp across the curve, we would expect the value of the swap to be \$69,253 in our favor. The change in the value of just the fixed leg of the swap in isolation given this move would be \$71,789 in our favor.
- Suppose we pay fixed on a \$100mm 5-year swap at a rate of 5.60%. If the swap curve were to rally 1 bp across the curve, we would expect the value of the swap to be \$40,237 against us. If the swap curve were to rally 10 bps across the curve, we would expect the value of the swap to be $10 \times \$40,237 = \$402,370$ against us. This will serve as a good approximation. However, inasmuch as the PV01s of a swap is not constant but changes with the level of swap rates, this amount will not be precise. For small perturbations of the swap curve the error will be modest, but for larger changes it can be significant. For example, if the swap curve were to rally 100 bps, we would estimate that the value of the swap would be $100 \times \$40,237 = \$4,023,700$ against us. However, from Table 2.2, we know that the value of the swap given a 100 bp rally will be \$4,185,000 against us. The difference between these two numbers is due to the fact that PV01s do not remain constant as swap rates move.

- Suppose that we receive on \$50mm 30-year swaps at a rate of 6.20%. If the swap curve were to rally 1 bp across the curve, the value of the swap will be $0.5 \times \$112,636 = \$56,318$ in our favor. That is, PV01s are scalable in accordance with the notional of the swap.
- Suppose we receive on a \$100mm 5-year swap, but instead of receiving at the par swap rate of 5.60%, we receive at the rate of 5.59%. In lieu of receiving the par rate and instead receiving 1 bp lower, we should expect to be compensated for this. In fact, using the PV Coupon 01 number for a 5-year swap, we see that if we receive 5.59%, we should expect to be paid an upfront fee of \$43,321. Note that the 5-year swap with the coupon of 5.59% is an off-market transaction (i.e., it is not an NPV = 0 transaction in and of itself). However, the combination of the upfront fee and the off-market swap constitutes an NPV = 0 transaction.
- Suppose we pay fixed on a 2-year swap on a notional of \$1bn. Suppose we would like to be paid an upfront fee of \$1mm in the terms of the swap and will adjust the fixed rate accordingly. We should pay an above market coupon in the swap. Since the PV Coupon 01 on \$100mm of a 2-year swap is \$18,858, this means that the PV Coupon 01 on \$1bn of a 2-year swap is $10 \times \$18,858 = \$188,580$. Thus, we should pay $\$1,000,000 / \$188,580 \approx 5.3$ bps above the par 2-year swap rate, or $5.11\% + 0.053\% = 5.163\%$.

Let's consider now one final example that makes use of the approximation to the PV01 of a swap:

- Suppose that the 30-year semi bond swap rate is 3.583%, as depicted in Table 1.1. Suppose further that a client is interested in paying on \$100mm of a 30-year swap in which the fixed rate is quoted annual money. As discussed in Section 1.2.1, the annual money equivalent is 3.563%. If the client and dealer make a mistake and forget to convert the rate and instead they deal at 3.583% annual money instead of 3.563% annual money, then the client will have paid a coupon that is $3.583\% - 3.563\% = 2.0$ bps too high. We can approximate the PV Coupon 01 in this swap by setting

$$\begin{aligned} \text{PMT} &= 1 \text{ bp} \\ n &= 30 \\ i &= 3.563\% \end{aligned}$$

and computing a present value of approximately 18.25 bps, or around \$182,500 on a notional of \$100mm. Since the coupon in this trade